

Julia Fancher

Syracuse University

jlfanche@syr.edu • 412-789-7764

RESEARCH EXPERIENCE

Research on tidal disruption events conducted under Dr. Eric Coughlin

- Ran PHANTOM simulations with and without self gravity to isolate its effects on tidal disruption event debris streams; results published as first author
- Characterized the predictions of the CN22 model and their implications on the interpretation of observed data; results published as second author
- Ran PHANTOM simulations to test the predictions of the CN22 model and create a library of realistic stars; results published as first author
- Currently using TDE simulations with realistic stellar structure to predict the percentage of jetted TDEs

PUBLICATIONS AND PRESENTATIONS

- *Predicting the Properties of the Fallback Rate from Tidal Disruption Events: Investigating the Maximum Gravity Model.* **Fancher, J.**, et al. 2025. *Astrophysical Journal*, 990, 2
- *The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type.* Bandopadhyay, A., **Fancher, J.**, et al. 2024. *Astrophysical Journal Letters*, 962, L2
- *On the relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.**, et al. 2023. *Monthly Notices of the Royal Astronomical Society*, 523, 2323
- *Investigating the Maximum Gravity Model.* **Fancher, J.** Presented at the ESAC workshop in Madrid 2025.
- *The relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.**, et al. Presented at the 243rd AAS conference 2024; undergraduate Chambliss finalist.
- *Tidal Disruption Events.* **Fancher, J.** Presented at the PHANTOM workshop in Melbourne 2024.
- *The relative importance of shocks and self-gravity in modifying tidal disruption event debris streams.* **Fancher, J.** Presented at the Syracuse University Undergraduate Research Fair, fall 2023.
- *A Controlled Set of Numerical Experiments of Stars Destroyed by Supermassive Black Holes.* **Fancher, J.** Presented at the Syracuse University Undergraduate Research Fair, spring 2023.

EDUCATION

Syracuse University, Syracuse, NY

expected May 2026

Bachelor of Science in Physics and Applied Mathematics, Minor in Logic

- Renee Crown University Honors Program

North Allegheny Senior High School, Pittsburgh, PA

June 2022

- GPA: 4.0/4.0 unweighted
- AP Calculus BC, Chemistry, Human Geography, Language and Composition, Literature and Composition, Physics 1, Physics 2, Physics C: Electricity and Magnetism, Physics C: Mechanics, US History, Psychology

TEACHING AND MENTORING

Physics Peer Coach, Syracuse University

January 2023-Present

- Support students in PHY 102, PHY 222, PHY 211, and PHY 365

Holden Observatory Tour Guide, Syracuse University January 2023-Present

- Lead tours of Holden Observatory, including operating the telescope for evening tours

Society of Physics Students Executive Board Member June 2023-May 2025

- Served as treasurer and secretary during the 2023-2024 and 2024-2025 academic years

Friends of Inkululeko volunteer June 2023-Present

- Headed “Options After Matriculation” project while in Makhanda, South Africa
- Assisted in entrepreneurship skills workshop in Makhanda, South Africa
- Raise awareness and involvement in Inkululeko’s mission

Pathway Prep tutor, Syracuse University March 2023-December 2024

- Tutor students in rural Mexico in SAT math

EXTRACURRICULAR ACTIVITIES

- Society of Physics Students member, 2022-Present
- Syracuse Marching Band, alto saxophone, 2023-Present
- Kappa Kappa Psi member, 2024-Present; hold Sargent-at-Arms position 2025-Present
- Saint Luke the Evangelist Catholic Church teen volunteer, 2016-Present
- North Allegheny High School Marching Band Flute Section Leader, 2021-2022
- Girl Scouts of Western Pennsylvania, 2012-2022
- Westinghouse Science Honors Institute student, 2020-2021
- Allegheny Intermediate Unit Astronomy Intern, 2019-2020

AWARDS AND HONORS

- Astronaut Scholar, 2024 & 2025
- Goldwater Scholar, 2024
- Euclid Prize, 2025
- Emerging Research Fellow, Syracuse University, undergraduate research grant, 2023-Present
- Syracuse Leadership Scholar, tuition grant, 2022-Present
- SU Physics Coaching Award, 2023
- College of Arts and Sciences Dean's List, all semesters
- National Merit Finalist, 2022
- United States Presidential Scholars Program Candidate, 2022
- Pennsylvania Higher Education Assistance Agency Certificate of Merit, 2022
- North Allegheny Band Scholar, 2019-2021
- Girl Scouts of America Silver Award, 2019

OTHER EMPLOYMENT

University 100 Tour Guide, Syracuse, NY June 2025 - Present

- Lead prospective student tours of Syracuse University

Trainer, Chick-fil-A, Wexford, PA July 2020-December 2024

- Train fellow team members with skills needed to run multiple positions in front and back of house; complete one project monthly to improve communication
- “Equip for Excellence” award for demonstrating initiative and strong work ethic